

Question 1

Design and develop a Reinforcement Learning framework for a ride-sharing company.

A ride-sharing company like Uber or Ola uses Reinforcement Learning (RL) to efficiently match drivers with passengers. The RL agent learns from previous ride requests and continuously improves driver allocation to reduce waiting time and improve customer satisfaction.

RL Framework

Agent

- Ride-sharing management system

Environment

- Drivers, passengers, roads, traffic conditions

State Space

- Driver location
- Passenger location
- Traffic condition
- Driver availability
- Time of day

Action Space

- Assign nearest driver
- Assign another available driver
- Wait for a nearby driver
- Cancel request

Reward Function

Positive Rewards:

- Passenger picked up quickly
- Short waiting time
- Driver reaches destination efficiently

Negative Rewards:

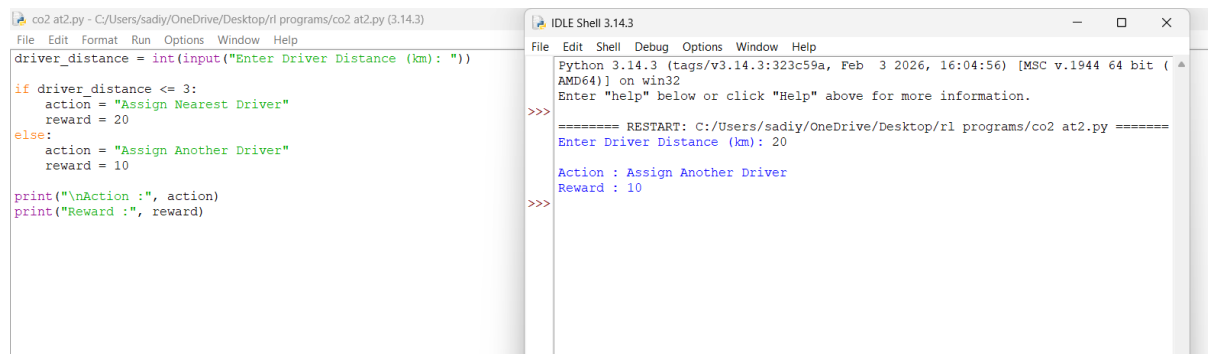
- Long waiting time

- Ride cancellation
- Driver travels unnecessary distance

Conclusion

The RL agent learns from every ride request and gradually improves driver allocation, reducing waiting time and improving service quality

Output:



The image shows a screenshot of a Python script and its execution. On the left, a text editor window titled 'co2_at2.py' contains the following code:

```
driver_distance = int(input("Enter Driver Distance (km): "))

if driver_distance <= 3:
    action = "Assign Nearest Driver"
    reward = 20
else:
    action = "Assign Another Driver"
    reward = 10

print("\nAction :", action)
print("Reward :", reward)
```

On the right, the IDLE Shell window shows the execution output:

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2_at2.py =====
Enter Driver Distance (km): 20

Action : Assign Another Driver
Reward : 10
>>>
```

Question 2

E-learning Platform Recommendation

An e-learning platform uses Reinforcement Learning to recommend personalized study materials based on student performance and interests.

RL Components

Agent

- Recommendation System

Environment

- Students and learning platform

States

- Student performance
- Completed lessons
- Quiz scores
- Learning progress

Actions

- Recommend easy topic
- Recommend advanced topic
- Suggest practice quiz

- Recommend video lecture

Rewards

Positive:

- Student completes lesson
- Higher quiz score
- Increased learning progress

Negative:

- Student skips lesson
- Low quiz score

Exploration

The system recommends new topics to discover student interests.

Exploitation

The system recommends topics that previously improved learning.

Conclusion

Balancing exploration and exploitation helps students learn effectively.

Output:

The screenshot displays a Python IDE with two windows. The left window shows the source code for a recommendation system, and the right window shows the execution output.

Source Code (co2 at2 2.py):

```
score = int(input("Enter Student Score: "))
if score >= 80:
    print("Recommendation : Advanced Lesson")
    print("Reward : +20")
else:
    print("Recommendation : Practice Quiz")
    print("Reward : +10")
```

Execution Output (IDLE Shell 3.14.3):

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Driver Distance (km): 20
Action : Assign Another Driver
Reward : 10
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Student Score: 95
Recommendation : Advanced Lesson
Reward : +20
```

Question 3

Robotic Vacuum Cleaner

A robotic vacuum cleaner uses Reinforcement Learning to clean rooms efficiently while avoiding obstacles.

RL Components

Agent

- Robot Vacuum Cleaner

Environment

- House floor

States

- Robot location
- Dirt level
- Obstacle position
- Battery level

Actions

- Move Forward
- Turn Left
- Turn Right
- Clean Area
- Return to Charging Station

Policy

The policy tells the robot which action to take in every situation.

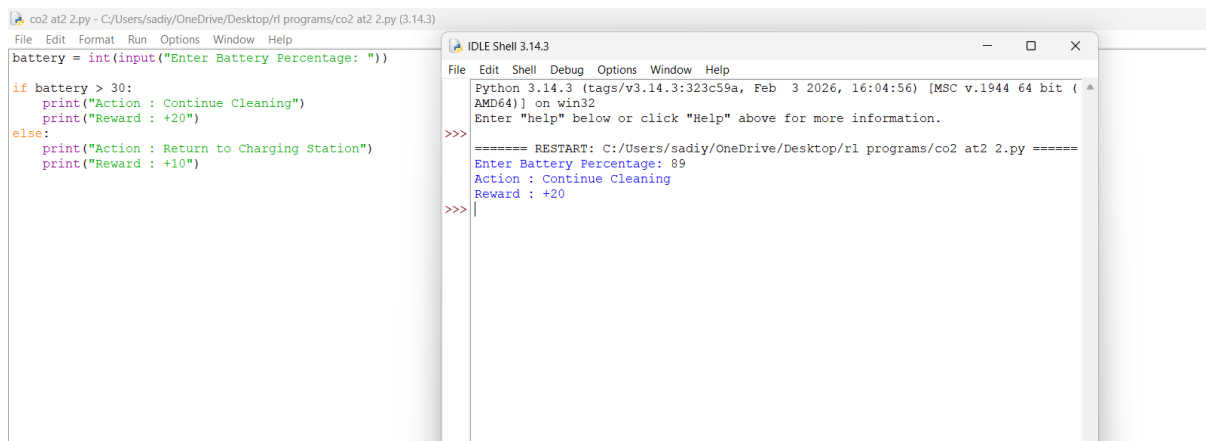
Value Function

The value function estimates the long-term benefit of each action. It helps the robot choose paths that clean more area while using less battery.

Conclusion

Over time, the robot learns better cleaning paths and avoids unnecessary movements.

Output:



```
co2 at2 2.py - C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py (3.14.3)
File Edit Format Run Options Window Help
battery = int(input("Enter Battery Percentage: "))
if battery > 30:
    print("Action : Continue Cleaning")
    print("Reward : +20")
else:
    print("Action : Return to Charging Station")
    print("Reward : +10")

IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Battery Percentage: 89
Action : Continue Cleaning
Reward : +20
>>>
```

Question 4

Cryptocurrency Trading System

A cryptocurrency trading system uses Reinforcement Learning to decide whether to buy, sell, or hold cryptocurrency based on market conditions.

RL Components

Agent

- Trading System

Environment

- Cryptocurrency market

States

- Current price
- Market trend
- Previous profit
- Trading volume

Actions

- Buy
- Sell
- Hold

Reward Function

Positive:

- Profit earned

Negative:

- Financial loss

Reward Design Challenges

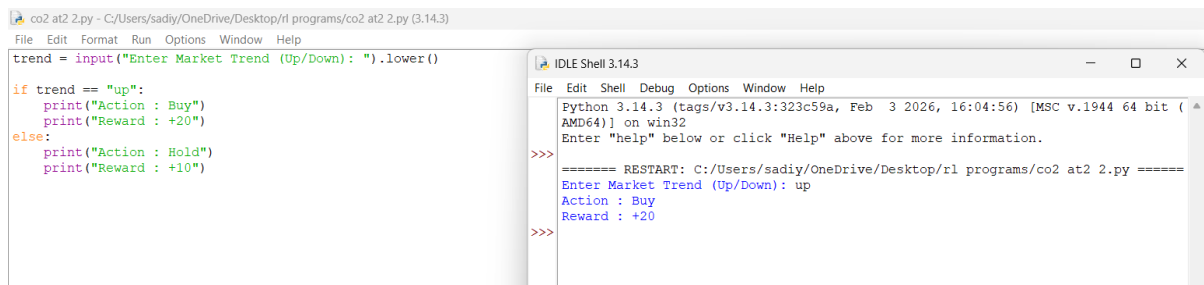
If the reward function is poorly designed:

- The agent may trade too often.
- It may take risky decisions.
- It may ignore long-term profit.

Conclusion

A well-designed reward function helps maximize profit while reducing financial risk.

Output:



The image shows a screenshot of a Python script and its execution output. The script is named 'co2_at2_2.py' and is located at 'C:/Users/sadiy/OneDrive/Desktop/rl programs/co2_at2_2.py (3.14.3)'. The script code is as follows:

```
trend = input("Enter Market Trend (Up/Down): ").lower()

if trend == "up":
    print("Action : Buy")
    print("Reward : +20")
else:
    print("Action : Hold")
    print("Reward : +10")
```

The output in the IDLE Shell 3.14.3 shows the following:

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2_at2_2.py =====
Enter Market Trend (Up/Down): up
Action : Buy
Reward : +20
```

Question 5

Smart Grid Electricity Distribution

A smart grid uses Reinforcement Learning to distribute electricity efficiently according to consumer demand while reducing energy wastage.

RL Framework

Agent

- Smart Grid Controller

Environment

- Electricity network

State Space

- Electricity demand
- Available power
- Battery level

- Time of day

Action Space

- Increase power supply
- Reduce power supply
- Store energy
- Use battery backup

Reward Mechanism

Positive Rewards:

- Stable electricity supply
- Low energy wastage
- Lower operating cost

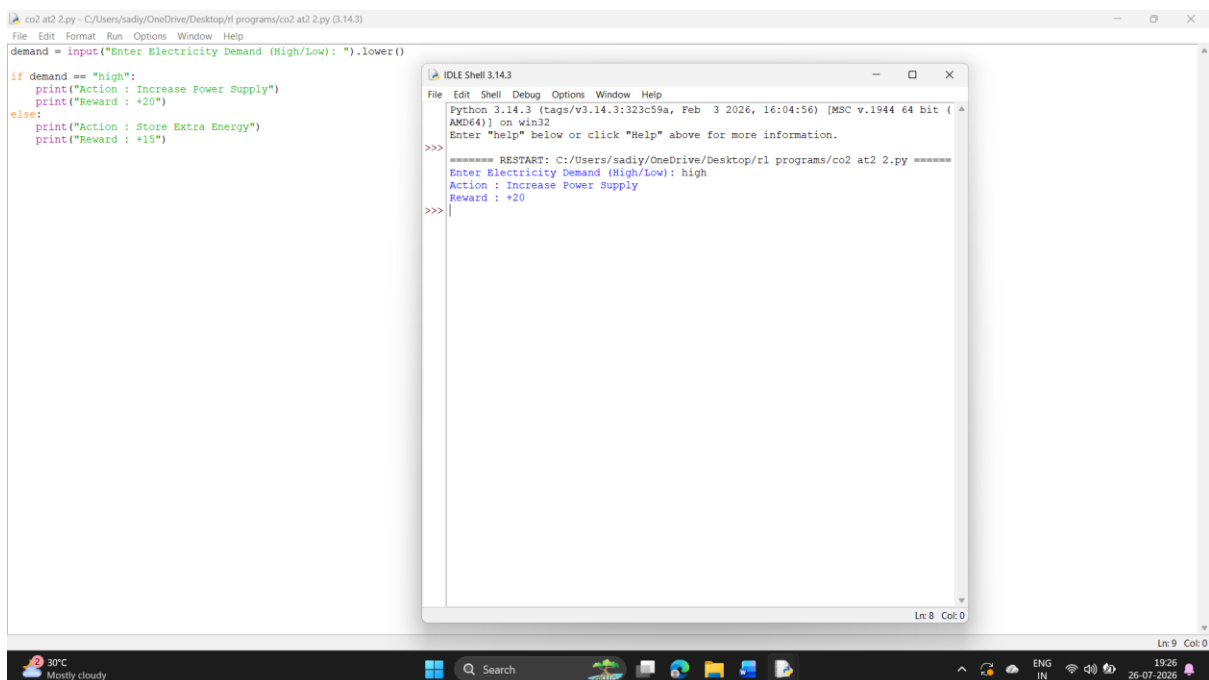
Negative Rewards:

- Power shortage
- Energy wastage
- High operating cost

Conclusion

The RL agent continuously learns electricity demand patterns and distributes power

Output:



The image shows a screenshot of a Windows desktop with two windows open. The background window is a Python IDLE editor showing a script named 'co2 at2 2.py'. The script prompts the user to enter electricity demand (High/Low) and then prints the corresponding action and reward. The foreground window is an IDLE Shell showing the execution of the script. It displays the prompt 'Enter Electricity Demand (High/Low): high', followed by the action 'Increase Power Supply' and the reward '+20'.

```
co2 at2 2.py - C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py (3.14.3)
File Edit Format Run Options Window Help
demand = input("Enter Electricity Demand (High/Low): ").lower()

if demand == "high":
    print("Action : Increase Power Supply")
    print("Reward : +20")
else:
    print("Action : Store Extra Energy")
    print("Reward : +15")

IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Electricity Demand (High/Low): high
Action : Increase Power Supply
Reward : +20
>>>
```

Question 6

Healthcare Monitoring System using Reinforcement Learning

A healthcare monitoring system uses Reinforcement Learning (RL) to adjust medication dosages based on a patient's health condition. The RL agent learns from patient responses and recommends the best dosage while reducing side effects.

RL Framework

Agent

- Healthcare Monitoring System

Environment

- Patient's body and medical condition

State Space

- Patient's blood pressure
- Heart rate
- Blood sugar level
- Previous medication dosage

Action Space

- Increase dosage
- Decrease dosage
- Keep dosage unchanged

Reward Function

Positive Rewards

- Patient's health improves
- Normal blood pressure
- Stable blood sugar level

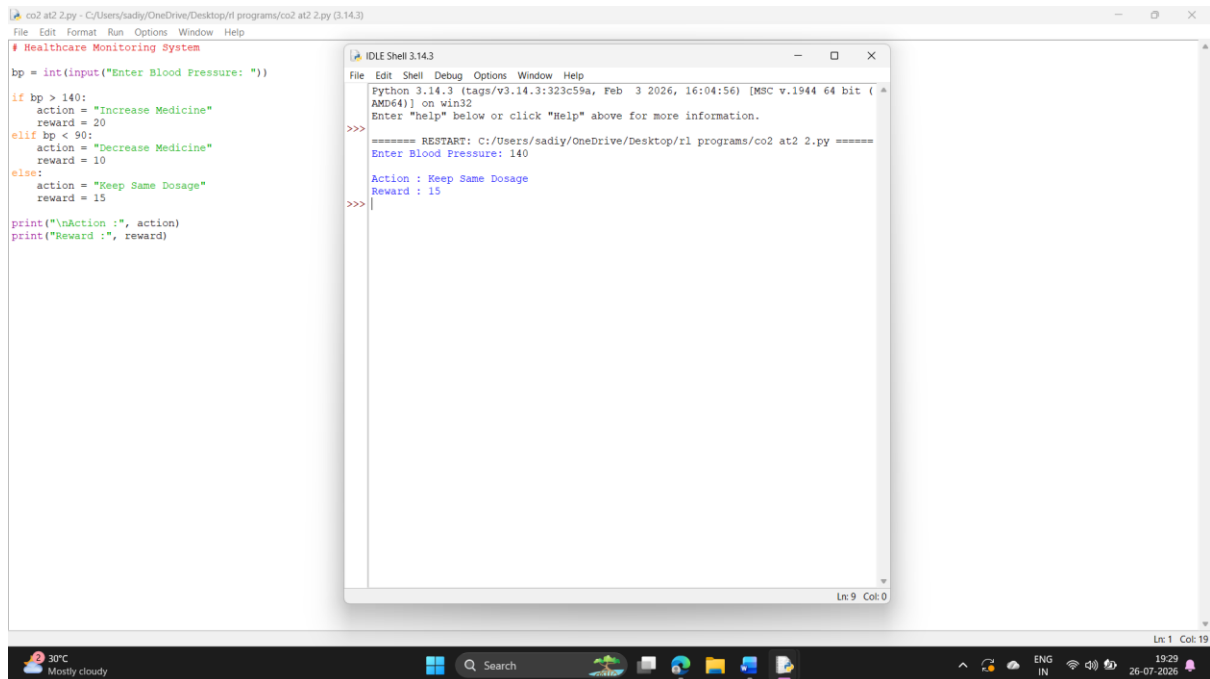
Negative Rewards

- Side effects occur
- Health condition worsens
- Incorrect dosage

Conclusion

The RL system learns from previous treatments and recommends the safest and most effective medication dosage.

Output:



The screenshot shows a Python IDE with a file named 'co2_at2_2.py'. The script is titled '# Healthcare Monitoring System' and contains logic for determining medication dosage based on blood pressure (bp). The logic is as follows:

- If bp > 140: action = "Increase Medicine", reward = 20
- elif bp < 90: action = "Decrease Medicine", reward = 10
- else: action = "Keep Same Dosage", reward = 15

The script prompts the user to 'Enter Blood Pressure: ' and prints the resulting action and reward. An output window titled 'IDLE Shell 3.14.3' shows the execution results:

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sady/OneDrive/Desktop/rl programs/co2_at2_2.py =====
Enter Blood Pressure: 140
Action : Keep Same Dosage
Reward : 15
>>>
```

Question 7

Drone Delivery Navigation

A drone delivery system uses Reinforcement Learning to find the shortest and safest route while avoiding obstacles.

RL Framework

Agent

- Delivery Drone

Environment

- Buildings, roads, weather, and obstacles

State Space

- Drone location
- Battery level
- Weather condition
- Distance to destination

Action Space

- Move Forward

- Turn Left
- Turn Right
- Return to Base

Exploration

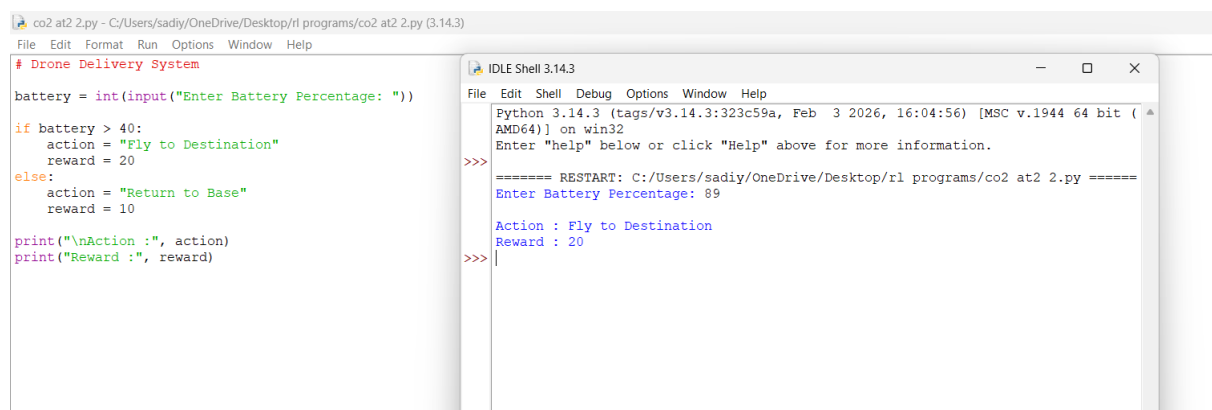
The drone tries different routes to discover shorter paths.

Exploitation

The drone uses the best route learned from previous deliveries.

Conclusion

Balancing exploration and exploitation helps the drone deliver packages quickly and safely.



The image shows a screenshot of a Python script titled 'co2 at2 2.py' and its execution in the IDLE Shell. The script is a simple reinforcement learning agent for a drone delivery system. It prompts the user to enter a battery percentage. If the battery is above 40%, it chooses the action 'Fly to Destination' and gives a reward of 20. Otherwise, it chooses 'Return to Base' and gives a reward of 10. The shell shows the script being executed, with the user entering '89' for the battery percentage. The output shows the action 'Fly to Destination' and the reward '20'.

```
# Drone Delivery System
battery = int(input("Enter Battery Percentage: "))
if battery > 40:
    action = "Fly to Destination"
    reward = 20
else:
    action = "Return to Base"
    reward = 10
print("\nAction :", action)
print("Reward :", reward)
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Battery Percentage: 89
Action : Fly to Destination
Reward : 20
>>>
```

Question 8

Manufacturing Robot Assembly

A manufacturing robot uses Reinforcement Learning to assemble products efficiently while reducing errors.

RL Framework

Agent

- Manufacturing Robot

Environment

- Factory assembly line

State Space

- Robot position
- Product status
- Tool availability

Action Space

- Pick Component
- Place Component
- Tighten Screw
- Inspect Product

Policy

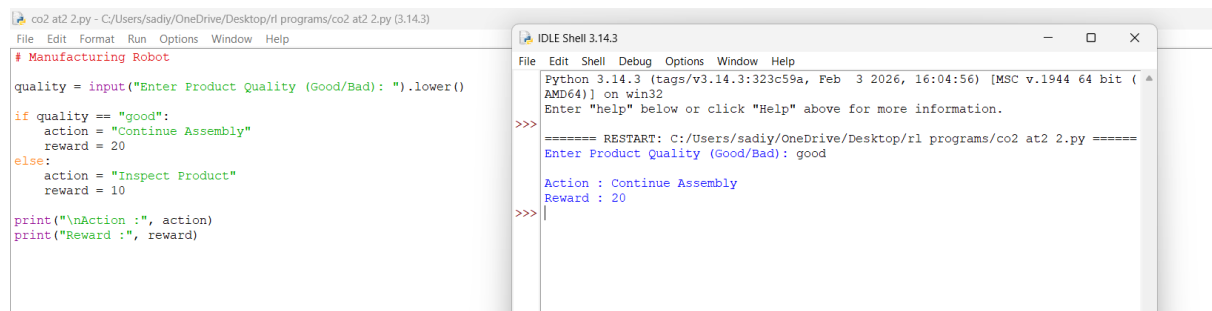
The policy tells the robot the best action to perform in each situation.

Value Function

The value function estimates which action gives the highest long-term reward. It helps the robot reduce assembly time and improve product quality.

Conclusion

The robot learns from experience and continuously improves assembly efficiency.



The image shows a screenshot of a Python script and its execution. On the left, a text editor window titled 'co2 at2 2.py' contains the following code:

```
# Manufacturing Robot
quality = input("Enter Product Quality (Good/Bad): ").lower()
if quality == "good":
    action = "Continue Assembly"
    reward = 20
else:
    action = "Inspect Product"
    reward = 10
print("\nAction :", action)
print("Reward :", reward)
```

On the right, an IDLE Shell window titled 'IDLE Shell 3.14.3' shows the execution output:

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Product Quality (Good/Bad): good
Action : Continue Assembly
Reward : 20
>>>
```

Question 9

Online Advertising System

An online advertising system uses Reinforcement Learning to display advertisements that are most likely to be clicked by users.

RL Framework

Agent

- Advertising Recommendation System

Environment

- Website visitors

State Space

- User age
- Search history
- Previous clicks

- Interests

Action Space

- Show Advertisement A
- Show Advertisement B
- Show Advertisement C

Reward Function

Positive Rewards

- User clicks the advertisement
- User purchases the product

Negative Rewards

- User ignores the advertisement
- Low click-through rate

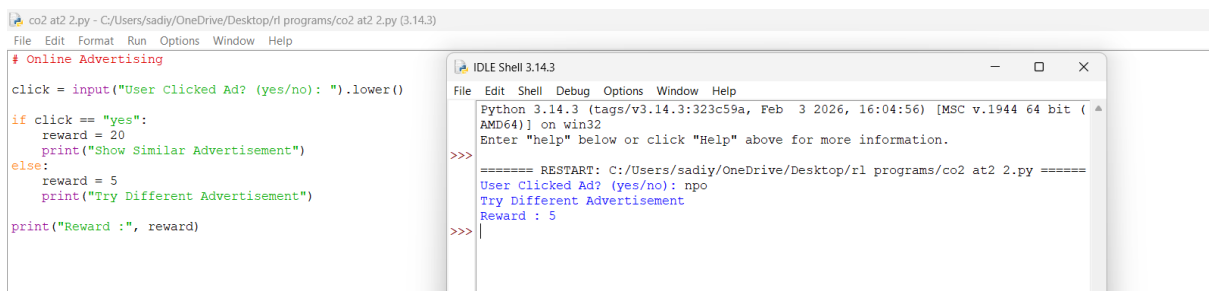
Reward Design Challenges

Poor reward design may:

- Recommend irrelevant advertisements
- Reduce click-through rate
- Decrease company revenue

Conclusion

A proper reward function helps the system recommend better advertisements and increase profits.



The image shows a Python script titled 'co2 at2 2.py' and its execution in the IDLE Shell. The script is for an online advertising system. It prompts the user to click on an advertisement (yes/no). If the user clicks 'yes', it awards a reward of 20 and suggests a similar advertisement. If the user clicks 'no', it awards a reward of 5 and suggests trying a different advertisement. The execution shows the user inputting 'npo' (no product) and receiving a reward of 5.

```

# Online Advertising

click = input("User Clicked Ad? (yes/no): ").lower()

if click == "yes":
    reward = 20
    print("Show Similar Advertisement")
else:
    reward = 5
    print("Try Different Advertisement")

print("Reward :", reward)

```

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
User Clicked Ad? (yes/no): npo
Try Different Advertisement
Reward : 5
>>>

```

Question 10

Smart Irrigation System

A smart irrigation system uses Reinforcement Learning to supply the correct amount of water based on soil and weather conditions.

RL Framework

Agent

- Smart Irrigation Controller

Environment

- Farm and crops

State Space

- Soil moisture
- Weather condition
- Water level
- Temperature

Action Space

- Turn Pump ON
- Turn Pump OFF
- Reduce Water Flow
- Increase Water Flow

Reward Function

Positive Rewards

- Healthy crops
- Water conservation
- Proper soil moisture

Negative Rewards

- Water wastage
- Dry soil
- Crop damage

Conclusion

The RL agent continuously learns environmental conditions and optimizes water usage, improving crop growth while conserving water.

co2 at2 2.py - C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py (3.14.3)

File Edit Format Run Options Window Help

```
# Smart Irrigation System

moisture = int(input("Enter Soil Moisture (%): "))

if moisture < 40:
    action = "Pump ON"
    reward = 20
else:
    action = "Pump OFF"
    reward = 15

print("\nAction :", action)
print("Reward :", reward)
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/co2 at2 2.py =====
Enter Soil Moisture (%): 95
Action : Pump OFF
Reward : 15
>>>
```